

Decentralized Training of Foundation Models in Heterogeneous Environments

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논문 정보

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- ▶ ETH Zürich, Switzerland/ Stanford University, USA/ Carnegie Mellon University/ Meta AI
- ▶ 36th Conference on Neural Information Processing Systems (NeurIPS 2022).

Heterogeneous network

- ▶ Challenge
 - ▶ Homogeneous 네트워크 환경에서만 foundation model 학습
- ▶ 모델의 크기 증가
- ▶ Heterogeneous network는 Homogeneous에 비해 search space가 큼
- ▶ Decentralized 환경은 communication bounded

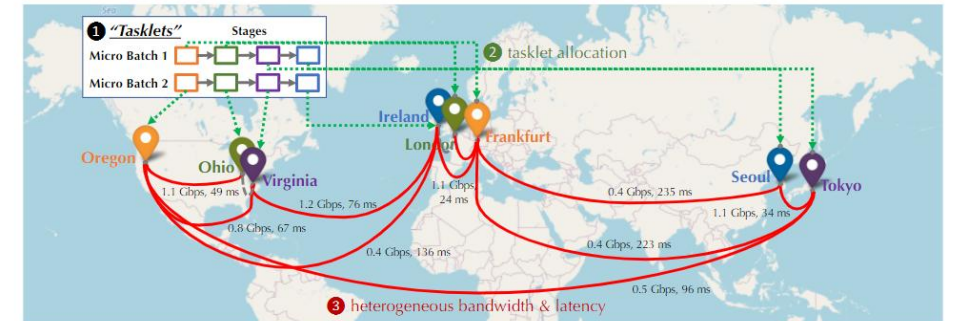


Figure 1: Given ① a set of computation tasklets involved in training foundation models (corresponding to different micro-batches and layers), and ② a heterogeneous network between devices, the goal is to find the optimal ③ allocation of tasklets to devices.

Approach

- ▶ Decomprising the cost model
 - ▶ Data parallelism cost : balanced graph partitioning
 - ▶ Pipeline parallelism cost: graph matching + traveling salesman
- ▶ Propose a novel scheduling algorithm
 - ▶ Tasklet배치 최적화

Formalization

- ▶ $D = \{d_1 \dots d_N\}$ N개 Devices 집합
- ▶ A: Network delay matrix, B: Network bandwidth matrix
 - ▶ d, d' 간의 delay: $\alpha_{d,d'}$, bandwidth: $\beta_{d,d'}$
 - ▶ 두 장치간 delay와 bandwidth는 대칭으로 간주 ($(d \rightarrow d' + d' \rightarrow d)/2$ 로 계산)
- ▶ $D_{DP} \times D_{PP} = N$
- ▶ $t_{i,j}$ ($i \in D_{DP}, j \in D_{PP}$)

$$\sigma \in \mathbf{D}^{D_{DP} \times D_{PP}}$$

$$\sigma^* = \arg \min_{\sigma \in \Sigma} \text{COMM-COST}(\sigma)$$

$t_{1,4}$	$t_{2,4}$	$t_{3,4}$	$t_{4,4}$
$t_{1,3}$	$t_{2,3}$	$t_{3,3}$	$t_{4,3}$
$t_{1,2}$	$t_{2,2}$	$t_{3,2}$	$t_{4,2}$
$t_{1,1}$	$t_{2,1}$	$t_{3,1}$	$t_{4,1}$

P_1

C_1

Modelling

$$\min_{\mathbf{C}_1 \dots \mathbf{C}_{D_{\text{rr}}}} \text{COMM-COST}(\mathbf{C}_1 \dots \mathbf{C}_{D_{\text{rr}}}) := \text{DATAP-COST}(\mathbf{C}_1 \dots \mathbf{C}_{D_{\text{rr}}}) \\ + \text{PIPELINEP-COST}(\mathbf{C}_1 \dots \mathbf{C}_{D_{\text{rr}}}) \quad (1)$$

► Data parallel

$$\text{DATAP-COST}(\mathbf{C}_j) = \max_{d \in \mathbf{C}_j} \sum_{d' \in \mathbf{C}_j - \{d\}} 2 \cdot \left(\alpha_{d,d'} + \frac{c_{\text{dp}}}{D_{\text{DP}} \beta_{d,d'}} \right) \quad \text{DATAP-COST}(\mathbf{C}_1 \dots \mathbf{C}_{D_{\text{rr}}}) = \max_{j \in [D_{\text{rr}}]} \text{DATAP-COST}(\mathbf{C}_j)$$

Modelling

► Pipeline parallel

$$\min_{\mathcal{M}} \max_{(d,d') \in \mathcal{M}} 2 \left(\alpha_{d,d'} + \frac{c_{FP}}{\beta_{d,d'}} \right)$$

$$\text{PIPELINEP-COST}(\mathbf{C}_1 \dots \mathbf{C}_{D_{PP}}) = \text{OPENLOOPTSP}(\hat{\mathbf{G}}_{\mathbf{C}_1 \dots \mathbf{C}_{D_{PP}}})$$

- $\mathbf{C}_j$ 와 $\mathbf{C}_{j+1}$ 에서 동일한 micro-batch를 갖는 device끼리만 통신한다.

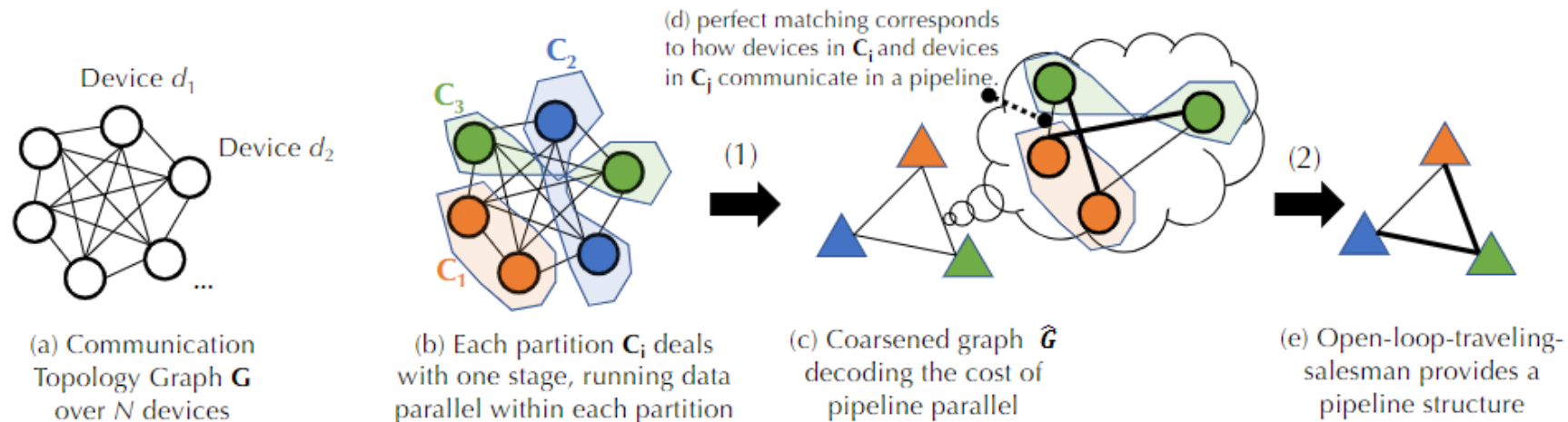


Figure 2: (a) Communication graph $\mathbf{G}$; and (b, c, d, e) an illustration of the cost model given $\mathbf{G}$.

Hybrid Genetic Algorithm

- ▶ 임의로 선택된 2 그룹의 랜덤한 노드끼리 교환하여 새로운 자식 파티션 o 생성
- ▶ o 를 기반으로 local search 수행하여 새로운 o^* 생성
- ▶ o^* 의 비용이 더 낮다면 다른 나쁜 자식을 제거

Improving Local Search Strategy

- ▶ 기존의 Local search 방식은 성능향상을 이뤄내지 못했다.
- ▶ 아래 2가지 관측을 통해 새로운 local search 전략 수립
 - ▶ C_j 에서 가장빠른 연결을 갖는 d_1 과 d_2 가 있을 때 d_1 을 제거해도 C_j 의 DP cost는 증가하지 않는다.
 - ▶ d_1 이 $C_{j'}$ 으로 이동하면 $d_1 \leftrightarrow d_2$ 연결이 빠르므로 C_j 와 $C_{j'}$ 간에 연결은 $d_1 \leftrightarrow d_2$ 이 될 것이다.

$$\text{GAIN}((d, C_j) \leftrightarrow (d', C_{j'})) = \frac{1}{|C_{j'}|} \sum_{d'' \in C_{j'}} w_{d_1, d''} - w_{d_1, d_2} + \frac{1}{|C_j|} \sum_{d'' \in C_j} w_{d'_1, d''} - w_{d'_1, d'_2}$$

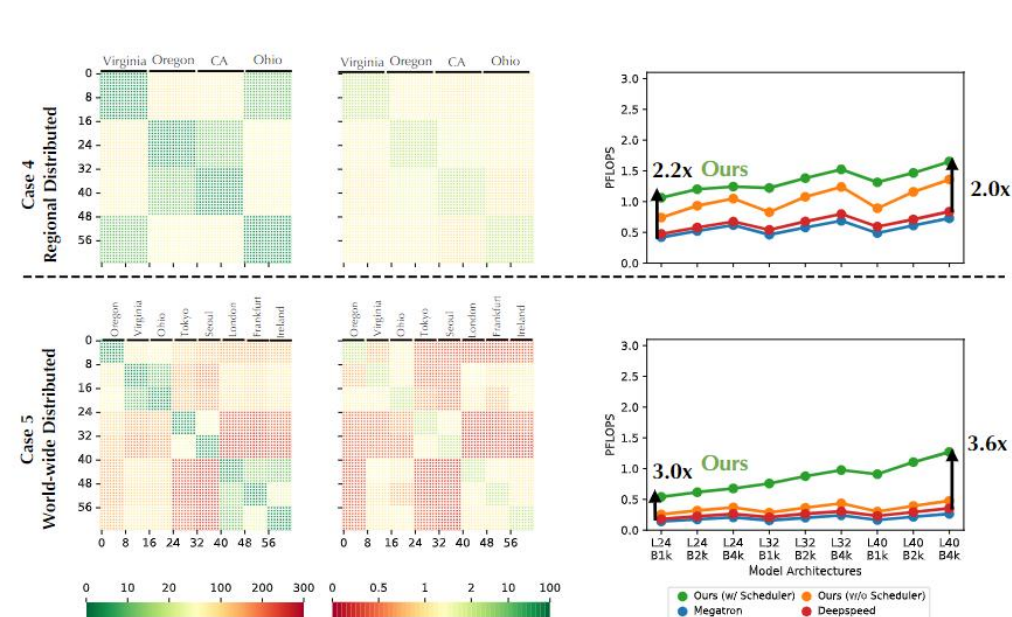
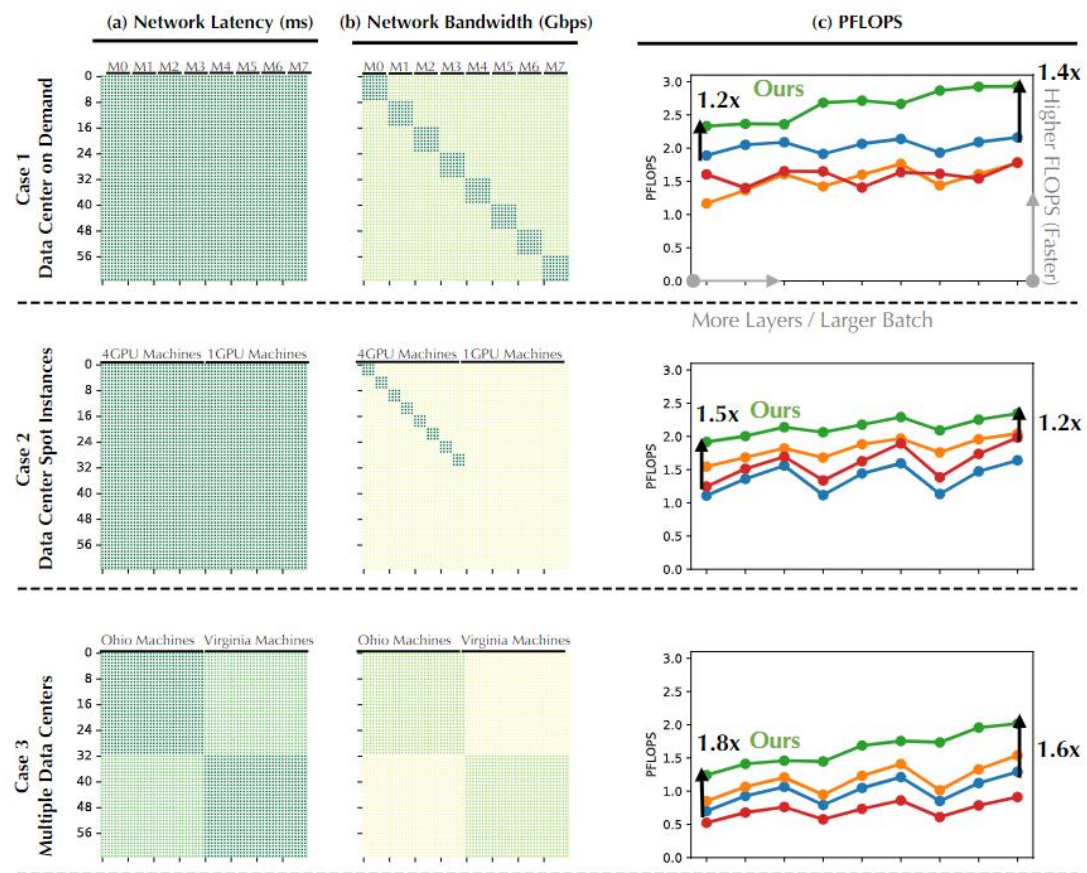
Experimental Setup

- ▶ V100 64 GPUs across AWS 8 regions
 - ▶ (Oregon, Virginia, Ohio, Tokyo, Seoul, London, Frankfurt, Ireland)
- ▶ Train GPT3.1-3B

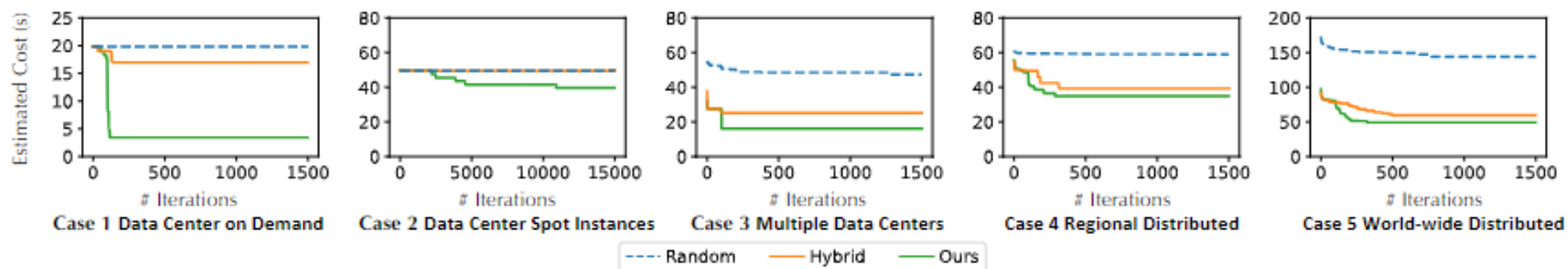
Cases

- ▶ Case 1. Data center on demand
 - ▶ 8 nodes(8 V100), intra node: NVLink 300GB/s, inter node: 25Gbps
- ▶ Case 2. Data center spot instances
 - ▶ 4 nodes(4 V100) & 32nodes(1 V100), intra node: 100Gbps, inter node: 10Gbps
- ▶ Case 3. Multiple Data Centers
 - ▶ 2 organizations (32 V100), 내부망: 10Gbps, 외부망: 1.12 Gbps / 10ms delay
- ▶ Case 4. Regional geo-distributed
 - ▶ 4 regions, 내부망: 2Gbps, 외부망: 1.0-1.3 Gbps / 10ms delay
- ▶ Case 5. World-wide geo-distributed
 - ▶ 8 regions, 내부망: 2Gbps / 5ms delay, 외부망: 0.3-1.3Gbps / 10-250ms delay

Evaluation



Evaluation



Result

- ▶ Case 1 our system is 1.7-3.5 X slower than Megatron/Deepspeed in case 1
- ▶ Network can be 100X slower
- ▶ In case 5, 4.8× speedup of Megatron, 3.6× speedup of Deepspeed

